Model-Based Investigation of Zn–CO_2 Electrochemical Systems for CO_2 Reduction to H_2 and Electricity Production

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Abstract

The global atmospheric CO_2 concentration increased to 420 ppm in 2023. This rise in CO_2 levels have led to several global challenges, including global warming, glacial melting, and sea level rise. Several CO_2 capture and utilization techniques are currently under development, including liquid-based methods (such as amine solutions, ionic liquids, alkaline solvents, and amino acids), solid-based approaches (including alkali metal oxides and carbonation), as well as membrane-based, electrochemical, and photochemical methods. However, the high costs associated with these technologies remain a significant barrier to their widespread implementation.

Recently, there has been a surge of interest in using electrochemical methods to convert CO_2 into useful products or fuels, offering a potential solution to the problem of rising global CO_2 emissions. When CO_2 dissolves in water, it reacts with water to form bicarbonate ions (HCO_3^-), carbonate ions (CO_3^{2-}), and hydrogen ions (H^+), thereby lowering the pH of the solution and creating an acidic environment. This acidic condition can be effectively utilized to drive hydrogen gas production at the cathode, providing a sustainable pathway for both CO_2 utilization and hydrogen generation. A schematic of a $Zn - CO_2$ system is shown in Fig. 1, where the acidity of the CO_2 saturated KOH solution is utilized for hydrogen gas generation on a Pt/C-coated carbon felt cathode. On the other hand, Zn metal is oxidized to form $Zn(OH)_4^-$ at the anode, while a cation exchange membrane separates the anode and cathode, allowing only K^+ ions to pass from the anode to the cathode. These potassium ions react with HCO_3^- to form $KHCO_3$ in the solution.

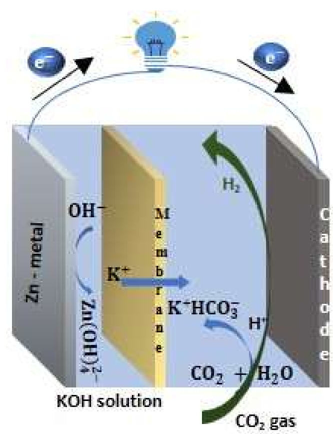

Figure 1: Schematic of Zn-CO2 electrochemical system

This study presents a 2D multiphase physics-based model of the $Zn - CO_2$ system using the Euler-Euler multiphase flow model, Dilute solution theory, and porous electrode theory. The model is simulated using COMSOL Multiphysics and validated using experimental data from Kim et al. [1]. Furthermore, analyses are conducted on gas-phase distribution, reactant species concentration profiles, electrolyte potential distribution, and the effects of parameters such as porosity, cathode–membrane distance, and chemical reaction rates to enhance system performance.

References

[1] Changmin Kim, et al., 2019, *Angewandte Chemie*, 131(28), 9606–9611.